

MMF1921: Operations Research

Project 2 (Summer 2025)

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1 Introduction

This project aims to create an automated asset management system that utilizes previous asset and factor return data to dynamically build and rebalance investment portfolios. Leveraging the concepts of multifactor modeling and portfolio optimization, we developed a trading algorithm that uses rolling estimation and semiannual balancing to adapt to shifting market conditions.

The primary objective is to implement a data-driven investment strategy that maintains a low average turnover rate while optimizing the out-of-sample risk-adjusted performance as determined by the Sharpe ratio. To achieve this, we applied a range of base factor models including OLS model, FF model, LASSO model and BSS model to estimate expected returns. The ensemble method is used to combine the outputs of all factor models to improve robustness and reduce model noise. we also developed a PCA-based model to extract latent sources of return variation and evaluate its effectiveness comparing with ensemble model. The estimates derived from both models are then incorporated into a Mean-Variance Optimization (MVO) framework for portfolio construction. To enhance constant updating and recalibration, every modeling component is integrated into a rolling window architecture.

We assess the performance of our trading strategy using the provided three datasets, and the strategy would be further evaluated on 2 unseen datasets during the competition stage. This report outlines the model construction methodologies, the model performance evaluation and the rationale for model selection.

2 Methodology

2.1 Data Processing

We need to process dataset provided to get asset return r_i which is given by:

$$r_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1 \quad (1)$$

2.2 Factor Models

For this project, we select OLS, FF, LASSO and BSS model as our base factor models, which aim to provide estimate of the expected excess return and covariance matrix of individual assets. Each factor model discloses the relationship between asset performance and the systematic risk exposure by applying regression model over a rolling window of past data. For the four factor models, each will offer a unique trade-off between complexity, interpretability and predictive power, while all models aim to provide forecasted return and covariance matrix to generate ensemble estimates used to construct efficient portfolios through optimization. We also build PCA-based model which provides estimates under the principle components selection framework.

2.2.1 OLS model

We use OLS model to estimate expected excess returns and covariance matrix for each asset at each rebalancing point. The goal is to fit a linear relationship between the excess return and a set of risk factors by minimizing the sum of squares of the prediction error. At each rebalancing point, the model is re-estimated using a 60-month rolling window of historical data.

Model Structure

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_i^\top \mathbf{F}t + \varepsilon_{i,t} \quad (2)$$

- $r_{i,t}$ is the return of asset i at time t ;
- $r_{f,t}$ is the risk-free rate at time t ;
- F_t is the vector of factor returns;
- α_i is the intercept term for asset i ;
- β_{ik} is the factor loading of asset i on factor k ;
- $\varepsilon_{i,t}$ is the residual term (idiosyncratic component) at time t .

Parameters $\hat{\alpha}_i$ and $\hat{\beta}_i$ Estimate

To estimate asset expected returns μ and covariance matrix $\mathbf{Q}$, we first need estimate $\hat{\alpha}_i$ and $\hat{\beta}_i$. First We can expand this formula:

$$\begin{aligned} r_i - r_f &= \alpha_i + \sum_{k=1}^8 \beta_{ik} F_k + \varepsilon_i \\ &= \alpha_i + \beta_{i1} F_1 + \beta_{i2} F_2 + \cdots + \beta_{i8} F_8 + \varepsilon_i \end{aligned} \quad (3)$$

$$= 1 \cdot \alpha_i + F_1^{(T)} \cdot \beta_{i1} + \cdots + F_8^{(T)} \cdot \beta_{i8} \quad (4)$$

Then we can construct a matrix $\mathbf{X}$, $\mathbf{X} \in \mathbb{R}^{T \times (K+1)}$

Where:

- T: Estimated window length (e.g. last 5 years = 60 months)
- K: Number of factors, in this project, K=8

$$R_i - R_f = \begin{bmatrix} 1 & F_1^{(1)} & F_2^{(1)} & \cdots & F_8^{(1)} \\ 1 & F_1^{(2)} & F_2^{(2)} & \cdots & F_8^{(2)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & F_1^{(T)} & F_2^{(T)} & \cdots & F_8^{(T)} \end{bmatrix} \cdot \begin{bmatrix} \alpha_i \\ \beta_{i1} \\ \beta_{i2} \\ \vdots \\ \beta_{i8} \end{bmatrix} + \begin{bmatrix} \varepsilon_i^{(1)} \\ \varepsilon_i^{(2)} \\ \vdots \\ \varepsilon_i^{(T)} \end{bmatrix} \quad (5)$$

We can get the matrix $\mathbf{X}$:

$$\mathbf{X} = \begin{bmatrix} 1 & F_1^{(1)} & F_2^{(1)} & \cdots & F_8^{(1)} \\ 1 & F_1^{(2)} & F_2^{(2)} & \cdots & F_8^{(2)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & F_1^{(T)} & F_2^{(T)} & \cdots & F_8^{(T)} \end{bmatrix} \quad (6)$$

Let introduce $y_i, y_i = r_i - r_f$, then based on equations (5) and (6), we can get that:

$$y_i = r_i - r_f = \mathbf{X} \cdot \beta_i + \varepsilon_i \quad (7)$$

where:

- $y_i \in \mathbb{R}^T$ is a vector of asset i 's excess return over T time periods.
- $X \in \mathbb{R}^{T \times (K+1)}$ is the matrix shown in equation (6).
- $\beta_i \in \mathbb{R}^{K+1}$ is the coefficient vector, and $\beta_i^\top = [\alpha_i, \beta_{i1}, \beta_{i2}, \dots, \beta_{iK}]$.
- $\varepsilon_i \in \mathbb{R}^T$ is the vector of random residuals.

For OLS model, it aims to minimize the sum of squares of the prediction error, then our objective function can be expressed by the equation that:

$$L(\beta_i) = \min_{\beta_i} \|y_i - X\beta_i\|^2 = (y_i - X\beta_i)^\top (y_i - X\beta_i) \quad (8)$$

And we can expand the equation (8), and get that:

$$L(\beta_i) = y_i^\top y_i - y_i^\top X\beta_i - (X\beta_i)^\top y_i + \beta_i^\top X^\top X\beta_i \quad (9)$$

$$= y_i^\top y_i - y_i^\top X\beta_i - y_i^\top X\beta_i + \beta_i^\top X^\top X\beta_i \quad (10)$$

$$= y_i^\top y_i - 2y_i^\top X\beta_i + \beta_i^\top X^\top X\beta_i \quad (11)$$

Then, to obtain $\hat{\beta}_i$, we take the derivative of the objective function $L(\beta_i)$ with respect to β_i and set it equal to zero, we can get that:

$$\frac{\partial L(\beta_i)}{\partial \beta_i} = -2X^\top y_i + 2X^\top X\beta_i = 0 \quad (12)$$

$$X^\top X\beta_i = X^\top y_i \quad (13)$$

Then we can get the estimate $\hat{\beta}_i$:

$$\hat{\beta}_i = (X^\top X)^{-1} X^\top y_i \quad (14)$$

Since we set $\beta_i^\top = [\alpha_i, \beta_{i1}, \beta_{i2}, \dots, \beta_{iK}]$, then we can get estimate $\hat{\alpha}_i$:

$$\hat{\alpha}_i = \hat{\beta}_i(1) \quad (15)$$

μ and Q Estimation

Now, we get the estimate $\hat{\alpha}_i$ and $\hat{\beta}_i$, then we can estimate expected returns μ and covariance matrix Q .

First, for equation $r_{i,t} - r_{f,t} = \alpha_i + \beta_i^\top \mathbf{F}t + \varepsilon_i, t$, we take expectations on both sides of the equation:

$$\mu_i = \mathbb{E}[r_{i,t} - r_{f,t}] = \mathbb{E} \left[\hat{\alpha}_i + \sum_{k=1}^K \hat{\beta}_{ik} F_{k,t} + \varepsilon_{i,t} \right] \quad (16)$$

$$= \hat{\alpha}_i + \sum_{k=1}^K \hat{\beta}_{ik} \cdot \mathbb{E}[F_k] \quad (17)$$

$$= \hat{\alpha}_i + \sum_{k=1}^K \hat{\beta}_{ik} \cdot \bar{F}k \quad (18)$$

Where $\bar{F}k$ is the sample mean over the past 60 months.

In order to estimate matrix covariance Q , first we construct a matrix $V \in \mathbb{R}^{K \times N}$:

$$V = \begin{bmatrix} \hat{\beta}_{11} & \hat{\beta}_{12} & \cdots & \hat{\beta}_{1N} \\ \hat{\beta}_{21} & \hat{\beta}_{22} & \cdots & \hat{\beta}_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \hat{\beta}_{K1} & \hat{\beta}_{K2} & \cdots & \hat{\beta}_{KN} \end{bmatrix} \quad (19)$$

- K : number of factors, here $K=8$.
- N : number of assets, here $N=20$, corresponding to the number of assets in the first training dataset.

We can express the sample mean $\bar{F}$ by the equation that:

$$\bar{F} = \frac{1}{T} \sum_{t=1}^T F_t \quad (20)$$

Then the sample covariance is given by:

$$F = \frac{1}{T-1} \sum_{t=1}^T (F_t - \bar{F})(F_t - \bar{F})^T \quad (21)$$

And we also need to account for residual covariance, first we can construct a matrix $E \in \mathbb{R}^{T \times N}$, which is given by:

$$E = \begin{bmatrix} \varepsilon_1^{(1)} & \cdots & \varepsilon_N^{(1)} \\ \varepsilon_1^{(2)} & \cdots & \varepsilon_N^{(2)} \\ \vdots & \ddots & \vdots \\ \varepsilon_1^{(T)} & \cdots & \varepsilon_N^{(T)} \end{bmatrix}$$

Then the residual covariance can be expressed by:

$$D = \text{cov}(\varepsilon) = \frac{1}{T-1} E^T E \quad (22)$$

Then the asset return covariance matrix $Q \in \mathbb{R}^{N \times N}$ is given by:

$$Q = V^T F V + D \quad (23)$$

Rolling Estimation

In order to adapt to changing market, we implement the regression estimation using a rolling window method for OLS model. Specifically, we re-estimate the OLS regression for all assets at each rebalancing point (every 6 months) over the 5 years historical data.

At each rebalancing point t , we calculate the regression model using observations from $[t-59, t]$ (the training window length is 60 months) and get estimates of coefficients $\hat{\alpha}_i^{(t)}$ and $\hat{\beta}_i^{(t)}$.

And the expected excess return is calculated as:

$$\hat{\mu}_{i,t+1} = \hat{\alpha}_i^{(t)} + \hat{\beta}_i^{(t)\top} \bar{\mathbf{F}}_t \quad (24)$$

Similarly, we calculate the sample covariance matrix over the 60-month training window.

Strengths and Weaknesses

1. Strengths

- Simple and interpretable;
- Implement all factors;
- Provides a useful benchmark for other factor models;

2. Weaknesses

- May have overfitting problem due to high dimensionality;
- May have multicollinearity problem among the factors;
- Potential noisy predictors due to the lack of factor selection;

2.2.2 FF model

The Fama–French three-factor model is a widely adopted asset pricing model that extends the Capital Asset Pricing Model (CAPM). This model uses a subset of the 8 factors, including size (SMB) and value (HML), and the market excess return (Mkt_RF). This model aims to explain variations in stock returns through systematic exposure to these three factors, which are grounded in empirical asset pricing literature. At each rebalancing point, the FF model will be re-estimated using a 60-month rolling window of historical data.

Model Structure

The model for each asset i over time periods $t = 1, \dots, T$ is expressed as:

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_{i1}F_1 + \beta_{i2}F_2 + \beta_{i3}F_3 + \epsilon_{i,t} \quad (25)$$

where:

- $r_{i,t}$: return of asset i at time t
- $r_{f,t}$: risk-free rate at time t
- $F_1 = \text{Mkt_RF}$, $F_2 = \text{SMB}$, $F_3 = \text{HML}$
- α_i : intercept term for asset i
- β_{ik} : exposure of asset i to factor k
- $\epsilon_{i,t}$: error term

Parameters $\hat{\alpha}_i$ and $\hat{\beta}_i$ Estimate

The parameter estimation for the Fama–French model is performed via Ordinary Least Squares (OLS) regression. For each asset $i = 1, \dots, n$, we define the excess return vector:

$$y_i = \begin{bmatrix} r_i(1) - r_f(1) \\ \vdots \\ r_i(T) - r_f(T) \end{bmatrix} \in \mathbb{R}^T \quad (26)$$

Let the factor matrix be:

$$X = \begin{bmatrix} 1 & F_{1,1} & F_{2,1} & F_{3,1} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & F_{1,T} & F_{2,T} & F_{3,T} \end{bmatrix} \in \mathbb{R}^{T \times 4} \quad (27)$$

Where each row corresponds to the intercept and the three Fama–French factors (Market, SMB, HML) at time t .

The coefficient vector $\hat{\beta}_i \in \mathbb{R}^4$ for asset i is estimated by solving the OLS problem:

$$\hat{\beta}_i = \arg \min_{\beta} \|y_i - X\beta\|_2^2 \quad (28)$$

Which has the closed-form solution:

$$\hat{\beta}_i = (X^T X)^{-1} X^T y_i = \begin{bmatrix} \alpha_i \\ \beta_{i1} \\ \beta_{i2} \\ \beta_{i3} \end{bmatrix} \quad (29)$$

Then we can get estimate $\hat{\alpha}_i$:

$$\hat{\alpha}_i = \hat{\beta}_i(1) \quad (30)$$

μ and Q Estimation

We first calculate the sample mean $\bar{F}_j$:

$$\bar{F}_j = \frac{1}{T} \sum_{t=1}^T F_j(t), \quad j = 1, 2, 3 \quad (31)$$

Then for each asset i , the expected return is given by:

$$\mu_i = \alpha_i + \beta_{i1}\bar{F}_1 + \beta_{i2}\bar{F}_2 + \beta_{i3}\bar{F}_3 \quad (32)$$

To estimate covariance matrix Q , we construct the matrix $V \in \mathbb{R}^{3 \times N}$, which is given by:

$$V = \begin{bmatrix} \hat{\beta}_{1,1} & \hat{\beta}_{2,1} & \cdots & \hat{\beta}_{N,1} \\ \hat{\beta}_{1,2} & \hat{\beta}_{2,2} & \cdots & \hat{\beta}_{N,2} \\ \hat{\beta}_{1,3} & \hat{\beta}_{2,3} & \cdots & \hat{\beta}_{N,3} \end{bmatrix} \quad (33)$$

Then we estimate the sample covariance matrix:

$$F = \frac{1}{T-1} \sum_{t=1}^T (\mathbf{F}_t - \bar{\mathbf{F}})(\mathbf{F}_t - \bar{\mathbf{F}})^T \in \mathbb{R}^{3 \times 3} \quad (34)$$

Specifically, for asset i , we first compute the residual time series:

$$\epsilon_{i,t} = y_{i,t} - X_t \hat{\beta}_i, \quad t = 1, \dots, T \quad (35)$$

where:

- $y_{i,t}$ is the excess return of asset i at time t
- X_t is the t -th row of the factor matrix X , i.e., $[1, F_{1,t}, F_{2,t}, F_{3,t}]$
- $\hat{\beta}_i$ is the estimated coefficient vector from OLS regression

We then compute the residual variance using an unbiased estimator that adjusts for degrees of freedom:

$$D = \sigma_{\epsilon_i}^2 = \frac{1}{T-d} \sum_{t=1}^T \epsilon_{i,t}^2 \quad (36)$$

where $d = 4$ is the number of estimated parameters in the FF model (intercept and 3 factors).

Estimate the covariance matrix of asset returns $Q \in \mathbb{R}^{n \times n}$ using a linear factor model approach,

which decomposes total asset return variability into systematic and idiosyncratic components:

$$Q = V^T FV + D \quad (37)$$

Rolling Estimation

To adapt to the dynamics of the market, we implement a rolling estimation framework for FF model. At each rebalancing point t (every 6 months), the model is re-estimated using the 5 years historical data.

- For each asset i , we construct the regression based on the observations from $[t - 59, t]$;
- Then we estimate the coefficients $\hat{\alpha}_i^{(t)}$ and $\hat{\beta}_i^{(t)}$;
- We then can get the estimate of expected excess return using the coefficients and we also calculate the covariance matrix Q using the same 60-month window;

Strengths and Weaknesses

1. Strengths

- Economically grounded factor selection: The FF model uses well-established factors (Mkt-RF, SMB, HML) that are motivated by asset pricing theory;
- Lower dimensionality due to limiting the model to three factors;

2. Weaknesses

- May overlook important factors due to limiting the model to three fixed factors;
- Residual risk may remain high due to leaving most of the asset return variation unexplained;

2.2.3 LASSO model

This formulation not only regularizes the coefficients but also performs variable selection by shrinking some factor loadings exactly to zero, thereby identifying a parsimonious model for each asset.

Model Structure

The LASSO (Least Absolute Shrinkage and Selection Operator) model augments the OLS regression with an ℓ_1 -norm penalty on the coefficients to promote sparsity. The model uses the same 8 factors as the OLS model, but automatically selects the most influential subset by shrinking insignificant coefficients toward zero.

For each asset i , the penalized regression is formulated as:

$$\min_{\beta_i} \|y_i - X\beta_i\|_2^2 + \lambda \|\beta_i\|_1 \quad (38)$$

Where:

- $\beta_i \in R^9$ are the coefficients.
- $\lambda > 0$ controls level of shrinkage.

Rolling Window Implementation

We implement the LASSO model with a rolling window framework to make it adaptive to realistic market conditions. At each rebalancing date t_k , the model is reconstructed with the most recent 60 months' data such that:

$$\tau_{t_k} = \{t_k - W + 1, \dots, t_k\} \quad (39)$$

The coefficient $\hat{\beta}_i^{(t_k)}$ is recalculated at each t_k .

Expected Return and Covariance Estimation

From the equation $\hat{y}_i = X\hat{\beta}_i$, we can derive the predicted return for each asset i which is given by:

$$\hat{y}_i = X\hat{\beta}_i\mu_i = \hat{\mu}_i = \frac{1}{T} \sum_{t=1}^T \hat{y}_{i,t} \quad (40)$$

For idiosyncratic risk, let ϵ denote residuals that:

$$\epsilon_i = y_i - X\hat{\beta}_i \quad (41)$$

Therefore, the idiosyncratic risk is:

$$Q = \frac{1}{T-1} E^T E \quad (42)$$

Then the total covariance matrix is given by:

$$Q = \frac{1}{2}(Q + Q^T) \quad (43)$$

Lambda Selection

λ is a regularization parameter in the LASSO model that controls the amount of shrinkage applied to the regression coefficients.

Specifically,

- Small λ makes the LASSO model similar to the OLS model, with most of the coefficients kept.
- Large λ effectively performs variable selection by forcing more coefficients toward 0.

We applied 4-fold-validation to determine the optimal λ value for the LASSO model. We tested λ values ranging from 0.001 to 0.1 by splitting the dataset into 4 folds and computing the average prediction error. The λ that minimizes the average error was selected as the optimal regularization parameter.

To ensure stability and reproducibility, we made the cross-validation process deterministic by keeping the original data in order without random shuffling. In our model, small λ values tend to perform better, indicating that the LASSO model closely resembled the OLS solution. This suggests that regularization offered limited benefit to the model because the full factor model was already sufficiently strong.

Strengths and Weaknesses

1. Strengths

- Automatic variable selection;
- Better generalization;
- Sparsity and interpretability

2. Weaknesses

- Too sensitive to lambda selection;
- Bias in coefficient estimates;
- Unstable variable selection

2.2.4 Best Subset Selection model

We use BSS model to dynamically identify the most appropriate and informative subset of explanatory factors that best explains each asset's historical monthly excess returns. By restricting the number of selected factors to a relatively small number K , BSS reduces model complexity and mitigate overfitting, thereby improving out-of-sample robustness. We reselect subsets for the model every 6 month, where we reapply the BSS procedure using a rolling historical window, enabling the model to adapt to changing market conditions. The resulting estimates of μ and Q are then used as inputs in the MVO framework to reconstruct the trading portfolio.

Model Structure

We applied a linear multi-factor regression model to estimate each asset's expected excess return. Let:

- $y_i \in \mathbb{R}^T$ denotes the observed monthly excess returns for stock i .
- $X \in \mathbb{R}^{T \times (p+1)}$ be the design matrix composed of the 8 factor returns and an intercept column, with the form:

$$X = \begin{bmatrix} 1 & F_1^{(1)} & F_2^{(1)} & \dots & F_8^{(1)} \\ 1 & F_1^{(2)} & F_2^{(2)} & \dots & F_8^{(2)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & F_1^{(T)} & F_2^{(T)} & \dots & F_8^{(T)} \end{bmatrix} \in \mathbb{R}^{T \times 9}. \quad (44)$$

- $\beta_i \in \mathbb{R}^{p+1}$ represent the coefficients for the intercept and factor exposures.
- $\epsilon_i \in \mathbb{R}^T$ denote the residuals of asset i .

The regression model is expressed as:

$$y_i = X\beta_i + \epsilon_i \quad (45)$$

Best Subset Selection Procedure

Rather than fitting the model with all 8 factors, we apply BSS to determine the subset of K that best explains the excess return of assets.

Specifically, for each feasible subset $S \subseteq \{1, \dots, p+1\}$ with $|S| = K$, we solve:

$$\min_{\beta_i} \|y_i - X\beta_i\|_2^2 \quad \text{subject to} \quad \|\beta_i\|_0 \leq K \quad (46)$$

For each subset:

- **Select the submatrix $X_s \in \mathbb{R}^{T \times |S|}$ corresponding to the chosen predictors.**
As X is the full designed matrix containing p factors and 1 intercept, $S \subset 1, 2, \dots, p+1$ would be the subset of column indices of the predictors that we choose. Such that X_s would be the submatrix of X composed when we only select columns that are in S .
- **Compute Ordinary Least Squares (OLS) estimate.**

$$\hat{\beta}_{i,S} = (X_s^T X_s)^{-1} X_s^T y_i \quad (47)$$

We repeat this process for each asset and obtain the final coefficient vector $\hat{\beta}$ by retaining the best-performing subset of each asset.

- **Compute the residual sum of squares (RSS).**
RSS measures the fitness of the model, which is the objective function that we want to minimize

for each asset. For asset i , with excess return r_i and predicted excess return $\hat{r}_i = X_S \hat{\beta}_{i,S}$, the RSS is computed with the formula:

$$RSS_S = \|y_i - X_S \hat{\beta}_{i,S}\|_2^2 \quad (48)$$

- **Select the best subset S^* with the lowest RSS**

As RSS measures the difference between the prediction and the true value, low RSS indicates that the model has a good fit. Therefore, selecting the subset that generates the lowest RSS ensures to have the best-fitting sparse model.

K Selection

In the BSS model, K is the maximum number of predictors allowed to enter the linear regression for each asset, controlling the sparsity level of the model. The choice of K is associated to the bias-variance trade-off. Specifically,

- A small value of K results in a highly sparse model which may lead to underfitting.
- A large value of K results in a nearly full model which may lead to overfitting.

To determine an appropriate value of K , we tested values from 1 to 5.

For each value of K , we applied BSS across all assets and recorded the in-sample R^2 . We observed that R^2 peaked at $K = 3$.

Therefore we chose $K = 3$ as the final setting.

Expected Return and Covariance Estimation

With the fitted model $\hat{y}_i = X \hat{\beta}_i$, the expected return is the average of predicted values μ_i would be expressed as the sample average of $\hat{r}_i$:

$$\mu_i = \frac{1}{T} \sum_{t=1}^T y_{i,t} = \frac{1}{T} \sum_{t=1}^T (X \hat{\beta}_i)_t \quad (49)$$

In matrix format,

$$\mu_i = \frac{1}{T} \begin{bmatrix} 1 & f_{1,1} & f_{1,2} & \cdots & f_{1,p} \\ 1 & f_{2,1} & f_{2,2} & \cdots & f_{2,p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & f_{T,1} & f_{T,2} & \cdots & f_{T,p} \end{bmatrix} \begin{bmatrix} \hat{\alpha}_i \\ \hat{\beta}_{i,1} \\ \vdots \\ \hat{\beta}_{i,p} \end{bmatrix} \quad (50)$$

Where:

- $\hat{\alpha}_i$ is the coefficient of intercept.
- $\hat{\beta}_{i,p}$ is the coefficient for factor p , in this problem $p = 8$.

Repeating this process for all the assets generates:

$$\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{bmatrix} \quad (51)$$

Where μ_i is the expected monthly excess return of asset i , predicted from the subset of selected factors.

Let ϵ_i be the residual vector, With the regression model, we could compute ϵ_i for each asset i:

$$\epsilon_i = y_{i,t} - \hat{y}_{i,t} = y_{i,t} - X\hat{\beta}_i \quad (52)$$

Where $T = 1, 2, \dots, T$.

Computing residual for all the assets, we could get the residual matrix E that:

$$E = [\epsilon_1 \epsilon_2 \dots \epsilon_n] \quad (53)$$

Incorporating the residual matrix, we could estimate the covariance matrix D using the sample covariance formula:

$$D = \frac{1}{T-1} E^T E = Cov [\epsilon_1 \epsilon_2 \dots \epsilon_n] \quad (54)$$

D captures the idiosyncratic risks not explained by the selected factors since each asset's return is modeled using different factors. And the covariance matrix is given by:

$$Q = V^T FV + D \quad (55)$$

$$Q = \frac{1}{2}(Q + Q^T) \quad (56)$$

Rolling Window and Rebalancing

To simulate realistic market conditions, we implement rolling window and rebalancing framework which separating the entire dataset into training window and live trading window. At each portfolio rebalancing point, we perform the following process:

For each rebalancing date t_k , we use the most recent 60 months of data to estimate the factor model for each asset. The rolling time index is:

$$\tau_{t_k} = \{t_k - W + 1, \dots, t_k\} \quad (57)$$

Historical window includes:

- Monthly excess return of each asset $r_i \in R^W$ where $i = 1, \dots, N$,
- Monthly returns of the 8 equity-based risk factors and the risk-free rate, combined into matrix X .

For each asset, we update the expected return:

$$\mu_i = \frac{1}{W} \sum_{t \in \tau_{t_k}} (X\hat{\beta}_i)_t \quad (58)$$

The covariance matrix is computed as:

$$Q = V^T FV + \frac{1}{W-1} E^T E \quad (59)$$

The portfolio is optimized and rebalance every 6 months and the resulting weights are kept until the next update. At each step, the window rolls forward by 6 months, allowing the model to incorporate the latest data while discarding outdated observations.

Strengths and Weaknesses

1. Strengths

- Sparse and interpretable;

- Custom factor structure;
- Reduce overfitting

2. Weaknesses

- Unstable factor selection;
- High computational cost

2.3 Ensembling of Factor Models for Portfolio Input Estimation

In order to improve the robustness and reduce the model noise, we apply ensemble method which combines the outputs of all factor models. We use the average expected return and covariance estimates from the four models instead of using single model's prediction. Then we use these ensemble estimates as inputs to the portfolio optimization.

First, we set $\hat{\mu}^{(m)} \in \mathbb{R}^N$ and $Q^{(m)} \in \mathbb{R}^{N \times N}$ indicate the expected return vector and the covariance matrix derived from the four factor models. Then the ensemble inputs are defined as:

$$\mu^{\text{ensemble}} = \frac{1}{M} \sum_{m=1}^M \hat{\mu}^{(m)} \quad (60)$$

$$Q^{\text{ensemble}} = \frac{1}{M} \sum_{m=1}^M Q^{(m)} \quad (61)$$

where $M = 4$ is the number of factor models.

This method helps us to reduce the individual weaknesses of each model and combine the strengths, leading to more stable and reliable decisions. Generally, the ensemble strategy outperforms the single model, with higher Sharpe ratios.

2.4 Principal Component Analysis Model

Principal component analysis (PCA) model is an unsupervised model that extract a set of orthogonal factors that capture the largest sources of variance in asset returns. Different from other factor models, this technique transforms correlated variables into a new set of uncorrelated variables called principle components, which are linear combination of the original variables. In this project, PCA method is used as an alternative factor model.

The procedure is as follows:

- Input Preparation

Let $F \in \mathbb{R}^{T \times p}$ denote the matrix of factor returns over T months, and we mean-center the data:

$$\tilde{F} = F - \bar{F} \quad (62)$$

Where $\bar{F}$ is the mean of each factor.

- Hyperparameters Selection

Instead of fixing the number of principle components based on an arbitrary variance benchmark, we choose to implement a rolling cross-validation method to get the optimal number of principle components under the dynamic selection framework.

Specifically, we divide the data into a training set (first 48 months), a testing set (next 6 months) and a purging window (3 months) over the 60 months rolling window. And the value of k which achieves the highest Sharpe ratio across all windows is selected as the number of principle components. In this project, we choose 5 principle components.

- Perform PCA

We compute the sample covariance matrix of the centered factor return matrix:

$$\Sigma = \frac{1}{T-1} \tilde{F}^\top \tilde{F} \quad (63)$$

Then we perform eigendecomposition on Σ :

$$\Sigma = V \Lambda V^\top \quad (64)$$

where:

- $V \in \mathbb{R}^{p \times p}$ contains the principal directions (eigenvectors).
 - $\Lambda \in \mathbb{R}^{p \times p}$ is a diagonal matrix of the corresponding eigenvalues (explained variances).
- Select principle components
- The number of principle components k is generated based on a cumulative variance threshold. And the selected principle components at time t is given as:

$$Z_t = \tilde{F}_t V_k \quad \text{for } t = 1, \dots, T \quad (65)$$

where $\tilde{F}_t$ is the mean-centered vector of factor returns at time t and V_k is matrix of the top k eigenvectors.

- Regression

We construct a design matrix which is given by:

$$X = \begin{bmatrix} 1 & Z_1 & \dots & Z_k \end{bmatrix} \quad (66)$$

We use the selected principle components to perform regression on each asset's excess return:

$$r_i = \alpha_i + \beta_i^\top Z + \varepsilon_i \quad (67)$$

- Estimate $\hat{\mu}$ and Q We calculate the mean and the covariance matrix of the selected principle variables:

$$\bar{Z} = \frac{1}{T} \sum_{t=1}^T Z_t \quad (68)$$

$$F_Z = \frac{1}{T-1} Z^\top Z \quad (69)$$

where $Z \in \mathbb{R}^{T \times k}$ is the matrix of top- k principal components obtained by projecting the factor returns onto the eigenvectors. Then the expected return for each asset i is given as:

$$\mu_i = \alpha_i + \beta_i^\top \bar{Z} \quad (70)$$

$$Q = \beta^\top F_Z \beta + D \quad (71)$$

where:

- β is the matrix of estimated factor loadings;

- F_Z is the sample covariance matrix of PCs;
- D is a diagonal matrix of residual variances.

$$Q = \frac{1}{2}(Q + Q^\top) \quad (72)$$

2.5 Portfolio Optimization (MVO)

2.5.1 Optimization Setup (Common to All Models)

For the purpose of optimizing portfolio, we intend to minimize portfolio risk while ensuring that:

- Expected portfolio return meets the target return.
- Portfolio is fully invested (weights sum up to 1).
- No short-selling.

Mean–Variance Optimization (MVO), originally proposed by Markowitz (1952), is a foundational technique in quantitative portfolio management. It formulates the trade-off between expected return and portfolio risk (measured by variance) as a constrained optimization problem. The objective is to identify a portfolio allocation that minimizes total risk while achieving a required level of expected return. In our implementation, we estimate the expected return vector $\mu \in \mathbb{R}^n$ and the covariance matrix $Q \in \mathbb{R}^{n \times n}$ by averaging the estimates from the four factor models (OLS, FF, LASSO and BSS) to get the ensemble estimates for the optimization, and we also use PCA method to get the estimates:

Let:

- w be the vector of portfolio weights where w_i represents the weight of asset i in the portfolio.
- μ be the ensemble vector of expected excess returns, computed as the average across the four factor models..
- Q be the ensemble asset return covariance matrix, obtained by averaging the covariance matrices from each model.

Such that, the linear program for the optimization problem is:

$$\begin{aligned} & \textbf{minimize} && w^T Q w \\ & \textbf{subject to} && \mu^T w \geq R_{\text{target}} \\ & && \mathbf{1}^T w = 1 \\ & && w \geq 0 \end{aligned}$$

The constraint $\mu^T w \geq R_{\text{target}}$ ensures that the portfolio achieves at least a minimum acceptable expected return, while the budget constraint $\mathbf{1}^T w = 1$ enforces full investment of capital. The no-short-selling constraint $w_i \geq 0$ for all i reflects practical trading limits and enhances solution stability. The target return R_{target} is set dynamically at the beginning of each investment year, and is defined as the mean of the market excess return over the 60 months rolling window:

$$R_{\text{target}} = \frac{1}{n} \sum_{i=1}^n \mu_i \quad (73)$$

This optimization is implemented using Python’s `cvxpy` package. At each rebalancing point t , we re-estimate μ and Q using the latest 60 months rolling window and then we get the ensemble estimates used to solve the MVO problem, we also use the outputs of PCA method as the inputs for MVO. The resulting portfolio weights are then held constant during the corresponding 1-year out-of-sample investment period.

3 Results

3.1 In-sample Analysis

	Dataset1	Dataset2	Dataset3
R^2	0.2282	0.5926	0.3325
Adj R^2	0.1358	0.5396	0.2581

Table 1: Ensemble

	Dataset1	Dataset2	Dataset3
R^2	0.1681	0.5767	0.2332
Adj R^2	0.1177	0.5370	0.2106

Table 2: PCA

3.1.1 R^2 Analysis

R^2 measures the proportion of return variance explained by the model.

$$R^2 = 1 - \frac{RSS}{TSS} \quad (74)$$

A high R^2 means that the model explains a considerable portion of the asset's return variance; while a low R^2 means that a lot of the variances are unexplained by the model. In addition, while a high R^2 is generally desirable, an excessively high value may indicate overfitting of the model, which is undesirable as it negatively impacts the model's generalization.

We computed R^2 for each model as part of the models' in-sample evaluation. OLS achieves the highest R^2 , indicating that it explains the largest portion of return variance among all models. That is because we fitted the model with all 8 factors with out regularization or variable selection. While this model may incline to overfitting.

We computed R^2 for both models across all three datasets, the ensemble model consists outperforms the PCA model, indicating stronger explanatory power. Specifically, the ensemble model achieves an average R^2 of around 0.3844 while the average R^2 for PCA model is approximately 0.3260. This suggests that by incorporating more factors, ensemble model captures a broader and more stable range of return drivers across different data environments.

3.1.2 Adjusted R^2 Analysis

Adjusted R^2 is a refinement of the R^2 which adjusts for the number of predictors, which is given as:

$$\text{Adjusted } R^2 = 1 - \left(\frac{RSS/(T - K - 1)}{TSS/(T - 1)} \right) \quad (75)$$

Adjusted R^2 makes it more suitable way to compare the performance of the models, especially when it is complex to evaluate models. A higher adjusted R^2 indicates that the model explains a larger proportion of the variance in the dependent variable, while accounting for the number of predictors used.

As shown in tables, the ensemble model consistently outperforms the PCA model with higher adjusted R^2 across all three datasets. Specifically, ensemble model achieves 0.1358, 0.5396 and 0.2581, while PCA yields 0.1177, 0.5370 and 0.2106, respectively.

These results illustrate that the ensemble model has stronger explanatory power since it can better capture asset-level variation in shifting marketing environments.

3.2 Out-of-sample Analysis

3.2.1 Sharpe Ratio and Turnover Rate Analysis

	Dataset1	Dataset2	Dataset3
Sharpe ratio	0.1695	0.0848	0.1989
Turnover	0.4574	0.5860	0.5054

Table 3: Ensemble

	Dataset1	Dataset2	Dataset3
Sharpe ratio	0.1625	0.0815	0.1734
Turnover	0.5907	0.6483	0.5727

Table 4: PCA

To evaluate the performance of each model beyond the in-sample calibration window, we compute the Sharpe ratio and the turnover rate for the ensemble strategy and PCA strategy.

Average excess return is computed by taking average of total portfolio excess return at time t :

$$\bar{R}_{po,t} = \frac{1}{T} \sum_{t=1}^T R_{po,t} \quad (76)$$

Where $R_{po,t}$ is the total portfolio excess return at time t .

Average volatility is computed as the sample standard deviation of annual excess returns:

$$\sigma_{po} = \sqrt{\frac{1}{T-1} \sum_{t=1}^T (R_{po,t} - \bar{R}_{po})^2} \quad (77)$$

Sharpe ratio serves to measure the excess return earned per unit of risk taken:

$$\text{Sharpe Ratio} = \frac{\bar{R}_{po}}{\sigma_{po}} \quad (78)$$

r_f has been omitted in the computation as the returns $\bar{R}_{po}$ have already been adjusted to be excess returns over the risk-free rate.

The turnover rate at time t is given by:

$$\text{Turnover}_t = \sum_i 1^N |w_{i,t}^{\text{new}} - w_{i,t-1}^{\text{old}}| \quad (79)$$

Where:

- $w_{i,t}^{\text{new}}$ is the weight of asset i at time t ;
- $w_{i,t-1}^{\text{old}}$ is the weight of asset i at time $t-1$;
- N is the total number of the assets.

As shown in the table 3 and 4, the ensemble model achieves higher Sharpe ratio in Dataset 1 (0.1695 vs 0.1625), Dataset 2 (0.0848 vs 0.0815), and Dataset 3 (0.1989 vs 0.1734), suggesting better risk-adjusted returns in all datasets compared to PCA model.

In terms of turnover, the ensemble model consistently yields lower turnover across the three datasets, indicating that it can provide more stable and cost-efficient strategy.

Therefore, the ensemble model generally delivers better risk-adjusted returns and lower turnover, making it more stable and cost-efficient strategy under changing marketing conditions.

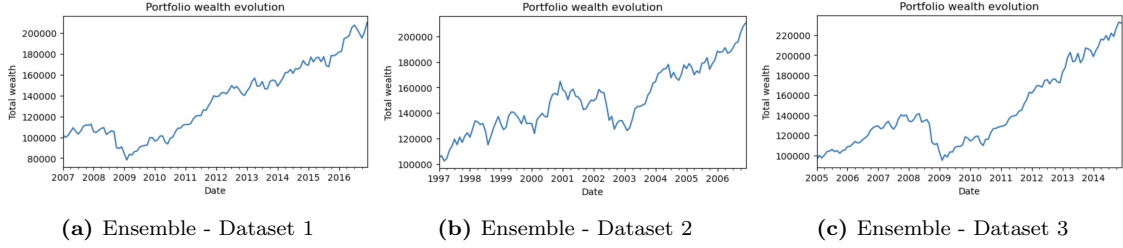


Figure 1: Portfolio Wealth Evolution Plots

3.2.2 Analysis of Portfolio Value over Periods

As shown in the portfolio wealth evolution plots, based on information in the 3 datasets, the portfolio value grew from \$100,000 to \$220,000 by 2014, achieving a cumulative return of more than 120%.

The smooth upward-trended curves (especially after 2010) show that the ensemble method has effectively reduced model-specific noise and overfitting, leading to improved out-of-sample robustness, demonstrating strong compound growth capability. In addition, though the portfolio value declined in the period 2008 to 2009, in the following years it demonstrated a rapid and strong recovery capacity, proving that the model remains resilient when faced with market shocks.

Across all 3 datasets, the ensemble model consistently shows a stable growth in the portfolio value despite of differences in time spans and market regimes. This consistency highlights the robustness and adaptability of the ensemble strategy, which may make it perform better in dynamic real-world settings.

3.2.3 Analysis of Composition of Portfolios

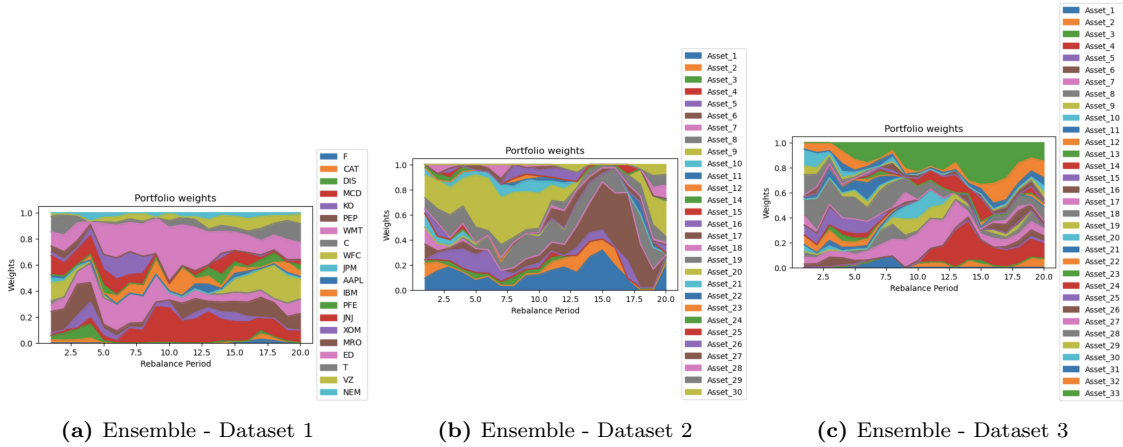


Figure 2: Portfolio Weights Plots

The figures illustrate the change of portfolio composition over 20 rebalancing periods for the ensemble strategy in different assets. We can derive from the figure that the portfolio remain continuously diversified, with most of the assets receiving non-zero allocation at different periods. At the same time, some assets are still allocated with small proportion, meaning that the portfolio has effective risk diversification without over-concentration on certain assets.

Although we can see some fluctuation in weights, particularly in mid-to-late periods, the overall evolution is smooth, suggesting that the model generates steady adjustments instead of abrupt

changes. So the ensemble method allows us to make gradual transitions by combining the outputs from the four models (OLS,FF,LASSO,BSS), which contributes to avoiding the overfitting problem. This will lead to a more robust and flexible portfolio which can adapt the changing market conditions while maintaining a balanced exposure to risk.

4 Conclusion

In this project we designed a dynamic portfolio construction and rebalancing system based on factor modeling and mean-variance optimization. We implemented 4 base return estimation model including OLS, Fama-French, LASSO, and Best Subset Selection and constructed an ensemble strategy by averaging the expected returns and covariances from the 4 basic models to enhance robustness and reduce model-specific risks. we also build a PCA-based factor model under the principle components selection framework. Through a rolling-window framework and semiannual rebalancing, we make the models more adapt to realistic market conditions. For three datasets, we evaluated both in-sample and out-of-sample performance of each model using metrics including R^2 , adjusted R^2 , Sharpe ratio, turnover rate as well as portfolio wealth evolution and portfolio weight plots, respectively, then subsequently identify the best performing model.

The ensemble model consistently achieves higher R^2 and adjusted R^2 across three datasets compared to PCA model, suggesting better in-sample fit. In terms of Sharpe ratio and turnover rate, the ensemble model generally outperforms the PCA model with overall higher Sharpe ratio and lower turnover rate, indicating better risk-adjusted return and cost-efficient strategy.

In conclusion, the ensemble model successfully integrated the advantages of individual models into a unified, balanced investment strategy demonstrating strong adaptability and practical viability in a dynamic market environment.